

Programming Fundamentals – Lab 06

Tasks

1. Student Attendance Counter (For Loop – Known Iterations)

A teacher wants to record attendance for 30 students in a class. Write a C program that uses a **for loop** to input attendance status (1 = Present, 0 = Absent) for each student and calculate the total number of students present.

Code:

```
int main(){
    int attendance;
    int totalpresent=0;
    for(int i=1;i<=10;i++){
        printf("enter student attendance = ",i);
        scanf("%d",& attendance);
        if(attendance==1){
            totalpresent++;
        }
    }
    printf("total present %d",totalpresent);
    return 0;
}
```

```
S:\Untitled15.exe
enter student attendance = 1
enter student attendance = 1
enter student attendance = 1
enter student attendance = 1
enter student attendance = 1
enter student attendance = 0
enter student attendance = 0
enter student attendance = 0
enter student attendance = 0
enter student attendance = 0
total present 5
-----
Process exited after 42.25 seconds with return value 0
Press any key to continue . . .
```

2. ATM Withdrawal System (While Loop – Condition Based)

An ATM allows a user to withdraw money until their balance becomes zero or negative. Write a C program using a **while loop** that keeps asking the user to enter the withdrawal amount and deducts it from the balance until the balance is exhausted.

Code:

```
int main() {
    int balance=12763, withdraw;

    while(balance > 0) {
        printf("Current balance: %d\n", balance);
        printf("Enter withdrawal amount: ");
        scanf("%d", &withdraw);

        balance = balance - withdraw;
```

```

    }

    printf("Balance exhausted!\n");
    printf("Final balance: %d\n", balance);

    return 0;
}

```

```

S:\Untitled15.exe
Current balance: 12763
Enter withdrawal amount: 32
Current balance: 12731
Enter withdrawal amount: 432
Current balance: 12299
Enter withdrawal amount: 234
Current balance: 12065
Enter withdrawal amount: 2
Current balance: 12063
Enter withdrawal amount: 12063
Balance exhausted!
Final balance: 0

-----
Process exited after 13.96 seconds with return value 0
Press any key to continue . . .

```

3. Password Retry System (Do-While Loop – At Least One Execution)

A system requires a user to enter a password. The correct password is 1234. Write a C program using a **do-while loop** that keeps asking the user to enter the password until the correct one is entered.

Code:

```

int main() {
    int input;

    do {
        printf("Enter password: ");
        scanf("%d", &input);
    } while (input != 1234);
}

```

```

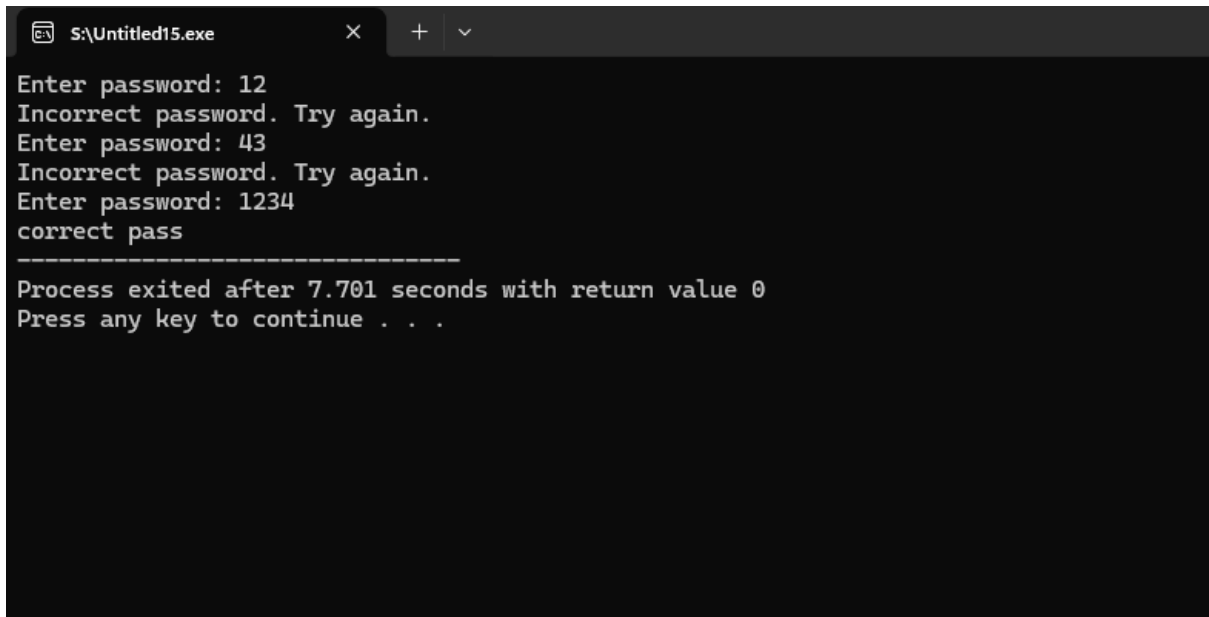
    if(input != 1234) {
        printf("Incorrect password. Try again.\n");
    }

} while(input != 1234);

printf("correct pass");

return 0;

```



```

S:\Untitled15.exe
Enter password: 12
Incorrect password. Try again.
Enter password: 43
Incorrect password. Try again.
Enter password: 1234
correct pass
-----
Process exited after 7.701 seconds with return value 0
Press any key to continue . . .

```

4. Sum of First N Natural Numbers (For Loop)

A user enters a number N. Write a C program that uses a **for loop** to calculate and display the sum of the first N natural numbers.

Code:

```

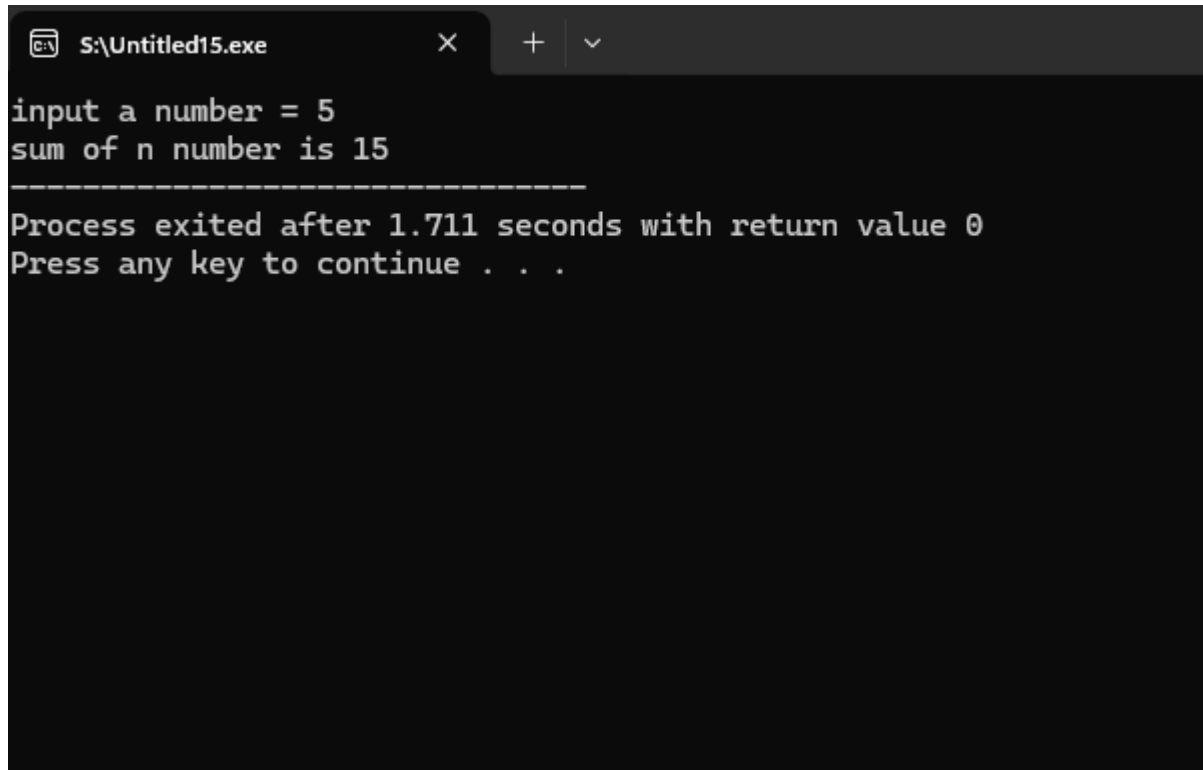
int main(){
    int n,sum=0;
    printf("input a number = ");
    scanf("%d",&n);
    for(int i=1;i<=n;i++){

```

```

        sum=sum+i;
    }
    printf("sum of n number is %d",sum);
    return 0;
}

```



```

S:\Untitled15.exe
input a number = 5
sum of n number is 15
-----
Process exited after 1.711 seconds with return value 0
Press any key to continue . . .

```

5. Temperature Monitoring System (While Loop – Dynamic Input Handling)

A weather station records temperature readings. The program should continuously accept temperature values until the user enters -999 (sentinel value). Use a **while** loop to calculate the average temperature of all valid inputs.

Code:

```

int main() {
    float temp, sum = 0;
    int count = 0;
    printf("Enter temp= ");
    scanf("%f", &temp);

    while (temp != -999) {
        sum =sum+ temp;

```

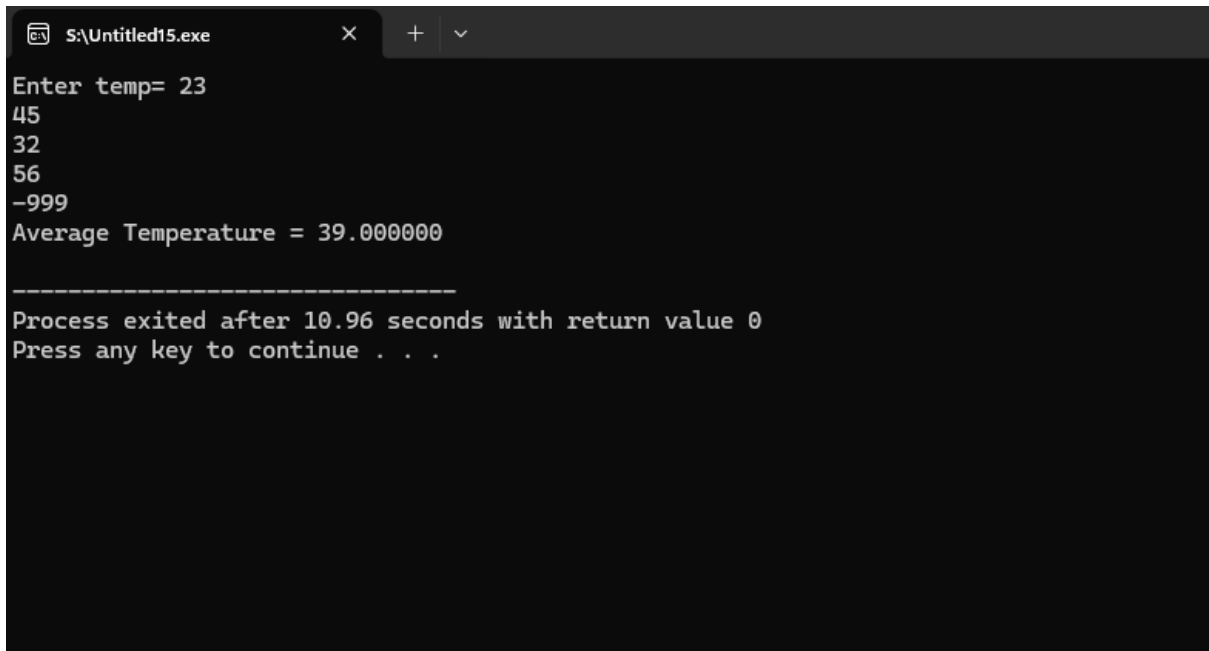
```

        count++;
        scanf("%f", &temp);
    }

    if (count > 0) {
        printf("Average Temperature = %f\n", sum / count);
    } else {
        printf("No valid temperature readings entered.\n");
    }

    return 0;
}

```



```

S:\Untitled15.exe
Enter temp= 23
45
32
56
-999
Average Temperature = 39.000000

-----
Process exited after 10.96 seconds with return value 0
Press any key to continue . . .

```

6. Multiplication Table Generator (For Loop – Controlled Iterations)

Write a C program that takes a number from the user and prints its multiplication table up to 10 using a **for loop**.

Code:

```

int main(){
    int n,table;

    printf("enter integer = ");

```

```

scanf("%d",&n);

for(int i;i<=10;i++){

    table=n*i;

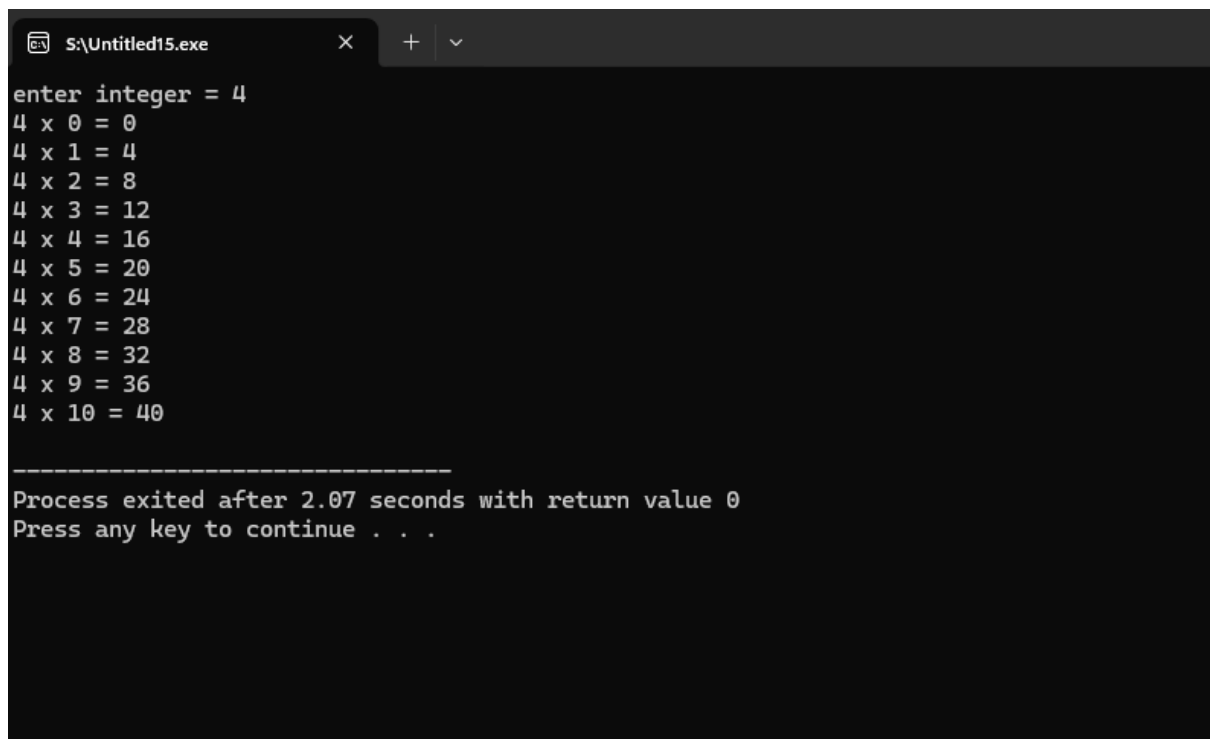
    printf("%d x %d = %d\n", n,i,table);

}

return 0;

}

```



```

S:\Untitled15.exe
enter integer = 4
4 x 0 = 0
4 x 1 = 4
4 x 2 = 8
4 x 3 = 12
4 x 4 = 16
4 x 5 = 20
4 x 6 = 24
4 x 7 = 28
4 x 8 = 32
4 x 9 = 36
4 x 10 = 40

-----
Process exited after 2.07 seconds with return value 0
Press any key to continue . . .

```

7. Menu-Driven Calculator (Do-While Loop – Exit Controlled Loop)

Design a simple calculator that shows a menu:

1. Addition
2. Subtraction
3. Exit

The program should continue to display the menu after each operation until the user chooses Exit. Use a **do-while loop** to implement this behaviour.

Code:

```
int main(){
    int i,n,input,result;

    do{
        printf("1.addition\n");
        printf("2.subtraction\n");
        printf("3.exit\n");
        printf("enter input= ");
        scanf("%d", &input);
        switch (input){
            case 1:
                printf("enter two numbers= ");
                scanf("%d %d",&i,&n);
                result=i+n;
                printf("result is %d\n", result);
                break;

            case 2:
                printf("enter two numbers= ");
                scanf("%d %d",&i,&n);
                result=i-n;
                printf("result is %d\n", result);
                break;

            case 3:
                printf("you choose exit");
                break;

            default:
                printf("invalid input try again\n");
```



```

    }
} while(input !=3);

return 0;

```

```

S:\Untitled14.exe
1.addition
2.subtraction
3.exit
enter input= 1
enter two numbers= 2
3
result is 5
1.addition
2.subtraction
3.exit
enter input= 2
enter two numbers= 3
5
result is -2
1.addition
2.subtraction
3.exit
enter input= 3
you choose exit
-----
Process exited after 20.91 seconds with return value 0
Press any key to continue . . . |

```

8. Number Guessing Game (While Loop – Control Flow Management)

Create a number guessing game where the correct number is 7. The user keeps entering guesses until they guess the correct number. Use a **while loop** and display appropriate messages ("Too High", "Too Low", "Correct!").

Code:

```

int main(){

    int n=7,guess;

    printf("input your guess= ");

    scanf("%d",&guess);

    while(guess!= n){

        if(guess>n){

            printf("guess is too high\n");

        }

        else if (guess<n){

            printf("guess is too low\n");

```

```

    }

    printf("guess again= ");

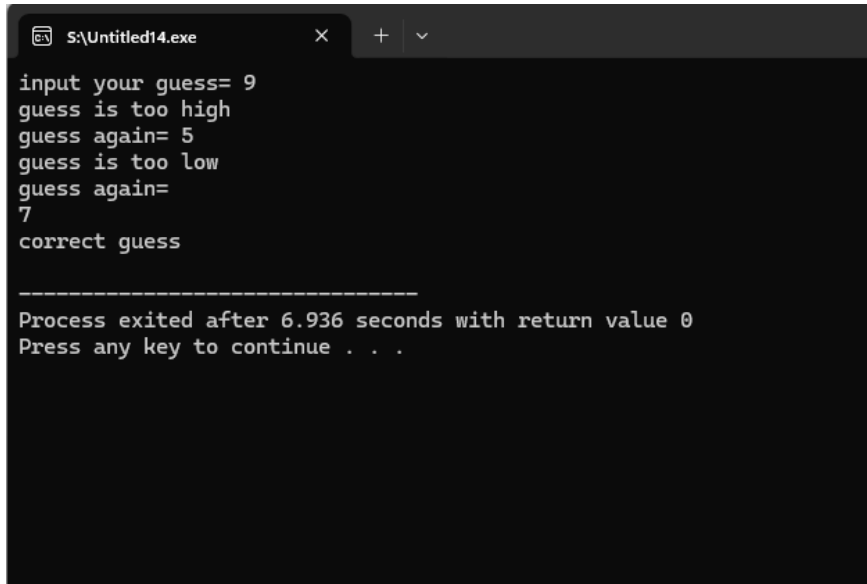
    scanf("%d",&guess);

}

printf("correct guess\n");

return 0;

```



```

S:\Untitled14.exe
input your guess= 9
guess is too high
guess again= 5
guess is too low
guess again=
7
correct guess

-----
Process exited after 6.936 seconds with return value 0
Press any key to continue . . .

```

9. Factory Production Planning System (For Loop – Logical Iteration & Multiplicative Growth)

A factory produces customized gift boxes. The manager receives an order to produce N boxes, where each box requires the number of decorative items equal to its box number.

For example:

- Box 1 requires 1 decorative item
- Box 2 requires 2 decorative items
- Box 3 requires 3 decorative items
- ... and so on until Box N

Write a C program that:

- Takes the number of boxes N as input
- Uses a **for loop** to calculate the total number of decorative items required
- Displays the final total

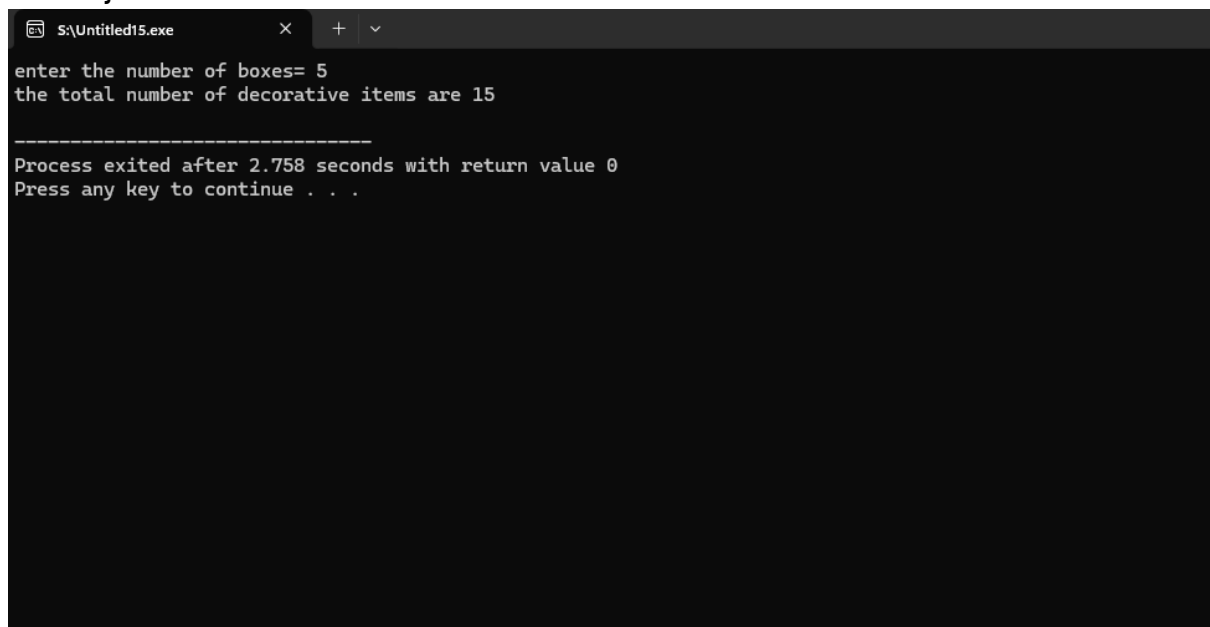
Code:

```
int main(){
```

```

    int n;
    int items=0;
    printf("enter the number of boxes= ");
    scanf("%d",&n);
    for(int i=1;i<=n;i++){
        items=items+i;
    }
    printf("the total number of decorative items are %d\n",items);
    return 0;
}

```



```

S:\Untitled15.exe
enter the number of boxes= 5
the total number of decorative items are 15

-----
Process exited after 2.758 seconds with return value 0
Press any key to continue . . .

```

10. University Data Analysis System (While Loop – Logical Classification & Sentinel Control)

A university is analyzing student test scores for research purposes. The system should continuously accept student scores (0–100).

The program must:

- Continue taking input until the user enters -1 (sentinel value)
- Count how many students scored:
 - **Distinction (≥ 75)**
 - **Pass (50–74)**
 - **Fail (< 50)**
- Display the total number of students in each category

Use a **while loop** to control the execution and ensure proper classification logic.

Code:

```

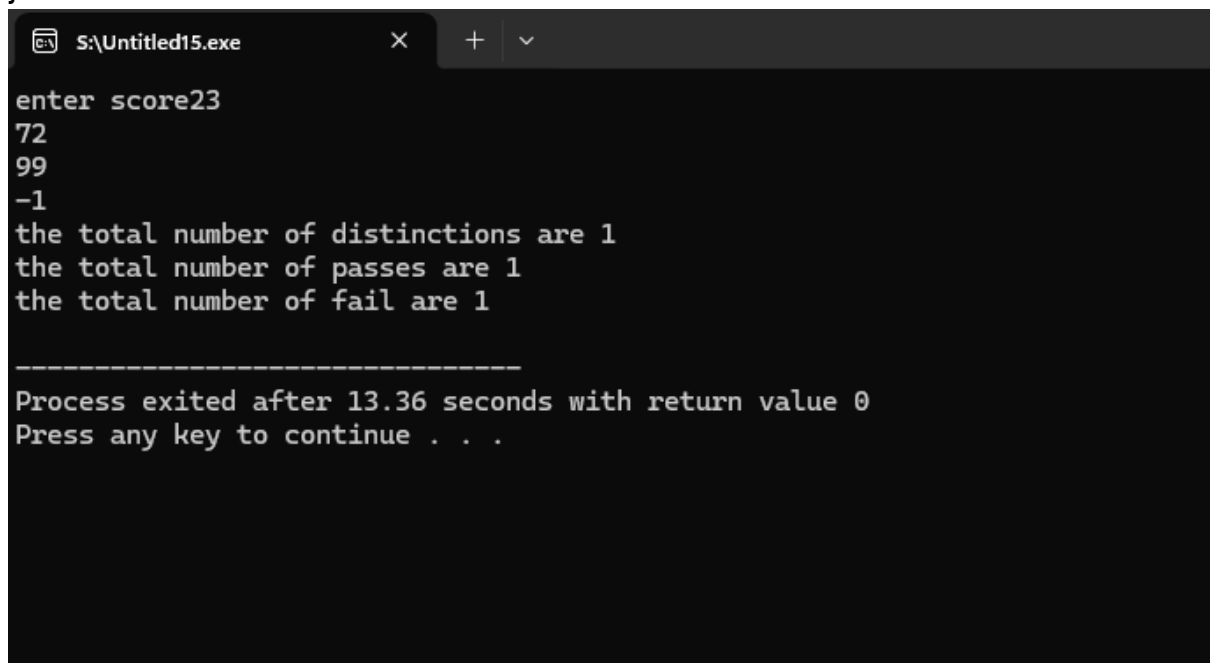
int main(){
    int score;

```

```

int distinction=0;
int pass=0;
int fail=0;
printf("enter score");
scanf("%d",&score);
while(score !=-1){
    if(score>=75){
        distinction++;
    }
    else if(score>50 && score<=74){
        pass++;
    }
    else if(score<50){
        fail++;
    }
    else{
        printf("invalid score");
    }
    scanf("%d",&score);
}
printf("the total number of distinctions are %d\n",distinction);
printf("the total number of passes are %d\n", pass);
printf("the total number of fail are %d\n", fail);
return 0;
}

```



```

S:\Untitled15.exe
enter score23
72
99
-1
the total number of distinctions are 1
the total number of passes are 1
the total number of fail are 1

-----
Process exited after 13.36 seconds with return value 0
Press any key to continue . . .

```

11. Online Shopping Cart Checkout System (Do-While Loop – Exit Controlled & User Choice Based)

An online shopping system allows customers to continuously add products to their cart.

Each time:

- The user enters the price of a product.
- The system adds it to the total bill.
- After each entry, the system asks whether the customer wants to add another product (1 = Yes, 0 = No).

The program must:

- Use a **do-while loop** so that at least one product is entered.
- Calculate the total bill amount.
- Apply a **5% discount** if the total bill exceeds 5000.
- Display the final payable amount.

Code:

```
int main(){
    float totalbill=0,price;
    int decision;
    do{
        printf("enter the price of product ");
        scanf("%f", &price);
        totalbill=totalbill+price;

        printf("do yo want to continue shopping [0 for NO and 1 for YES]");
        scanf("%d", &decision);
    } while(decision==1);
    if(totalbill>5000){
        totalbill=totalbill -(totalbill*0.05);
    }
    else{
        totalbill=totalbill;
    }
    printf("the final amount payable is %f",totalbill);
    return 0;
}
```

```
S:\Untitled15.exe
enter the price of product 4478
do yo want to continue shopping [0 for NO and 1 for YES]1
enter the price of product 9898
do yo want to continue shopping [0 for NO and 1 for YES]1
enter the price of product 64846
do yo want to continue shopping [0 for NO and 1 for YES]0
the final amount payable is 75260.898438
-----
Process exited after 16.76 seconds with return value 0
Press any key to continue . . .
```

12. Bus Seat Reservation Monitoring System (While Loop – Conditional Termination & Logical Control)

A bus has **40 seats** available. Passengers arrive one by one to reserve a seat. The system should:

- Allow seat booking until either:
 - All 40 seats are booked, OR
 - The operator enters 0 to stop reservations early.
- After each booking, display the number of remaining seats.
- Prevent booking if seats are already full.

Write a C program using a **while loop** to manage this reservation process and ensure proper condition control.

Code:

```
int main(){
    int seat=40,x=1;
    while(seat>0 ||x==0 ){
        printf(" if you want to rserve a seat press 1 (0 to stop)");
        scanf("%d",&x);
        if(x==1){
            seat--;
            printf("your seat is reserved\n");
            printf("%d seats are remaining\n", seat);
            if(seat==0){
                printf("sorry all the seats are alreaady reserved\n");
            }
        }
    }
}
```

```

        else if(x==0){
            printf("reservation stopped\n");
        }
        else{
            printf("invalid input choose 1 or 0");
        }
    }
    return 0;
}

```

```

S:\Untitled15.exe
if you want to rserve a seat press 1 (0 to stop)1
your seat is reserved
39 seats are remaining
if you want to rserve a seat press 1 (0 to stop)1
your seat is reserved
38 seats are remaining
if you want to rserve a seat press 1 (0 to stop)1
your seat is reserved
37 seats are remaining
if you want to rserve a seat press 1 (0 to stop)1
your seat is reserved
36 seats are remaining
if you want to rserve a seat press 1 (0 to stop)0
reservation stopped
if you want to rserve a seat press 1 (0 to stop)

```